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Ham Huang

Curriculum Vitæ[1]

Education

2023-Now **Ph.D. Psychology**,
Research on Computational Cognitive Science of Collective Cognition.
Advisors: Natalia Vélez (G1-3), Thomas L. Griffiths (G1-5), Diana Tamir (G4-5)
Princeton University.

2016-2020 **B.A. Philosophy, Minor in Logic**, Both GPA: 3.95/4.
B.A. Applied Mathematics, *Cluster in Computational Statistics*, GPA: 4/4.
B.A. Psychology, *Highest Honors*, GPA: 4/4.
Thesis: *A role for working memory in shaping the action policy for reinforcement learning*
Thesis Advisor: Anne G.E. Collins, Samuel D. McDougle
University of California, Berkeley, *Highest Distinction*, Cumulative GPA: 3.95/4.

Summer Schools

2024-2026 **Artificial Intelligence in Collectives**, *researcher network between Princeton and Tokyo University*.

2022 **Sloan-Nomis Summer School on the Cognitive Foundations of Economic Behavior**.

2022 **The Computational Summer School on Modeling Social and Collective Behavior**.

Journal Publications

Ham, H., Russek, E., Griffiths, T. L., & Vélez, N. (2027). Working memory capacity guides the division of cognitive labor. *In prep.*

Fascendini, B., Zhao, B., **Ham, H.**, & Vélez, N. (2027). Two-year-olds are intrinsically motivated to explore their competence. *In prep.*

Defendini, A., **Ham, H.**, Berkay, D., & Jenkins, A. C. (2026). Contributions of the brain's mentalizing network to inferences about others' stable and transient beliefs and preferences. *Journal of Cognitive Neuroscience*.

Ham, H., Zhao, B., Griffiths, T. L., & Vélez, N. (2025). Teaching recombinable motifs through simple examples. *Cognitive Science*.

Ham, H. & Jenkins, A. C. (2025). Social inequity disrupts reward-based learning. *Communications Psychology*.

Ham, H., McDougle, S. D., & Collins, A. G. E. (2025). Dual effects of dual-tasking on instrumental learning. *Cognition*.

[1]Current as of August, 2026. Publish as Huang Ham. * indicates equal contribution.

Plate, R. C., **Ham, H.**, & Jenkins, A. C. (2023). When uncertainty in social contexts increases exploration and decreases obtained rewards. *Journal of Experimental Psychology: General*.

Peer-reviewed Conference Publications

Ham, H., Correa, C., Bonan, Z., Griffiths, T. L., & Vélez, N. (2027). Emergence of abstractions in cultural transmission. *In prep*.

Ham, H., Griffiths, T. L., & Vélez, N. (2027). Dual eye-tracking reveal memory collaboration. *In prep*.

Ham, H., Weinhardt, D., Griffiths, T. L., & Vélez, N. (2026). Automated discovery of cognitive collaboration dynamics. *9th Annual Conference on Cognitive Computational Neuroscience*.

Ham, H., Arumugam, D., Correa, C., Bonan, Z., Griffiths, T. L., & Vélez, N. (2026). Rational teachers lie to bounder learners. *Proceedings of the Annual Meeting of the Cognitive Science Society, 48*.

Ham, H.*, Wang, M.*, Kable, J. W., & Jenkins, A. C. (2026). Modeling context-sensitive effects of inequity on reinforcement learning. *Proceedings of the Annual Meeting of the Cognitive Science Society, 48*.

Ham, H. & Jenkins, A. C. (2023). A computational model of lexical false memory based on semantic distance from word embeddings. *Proceedings of the Annual Meeting of the Cognitive Science Society, 45*.

Ham, H., Grahek, I., Bustamante, L., Daw, N., Caplin, A., & Musslick, S. (2022). Leveraging psychometrics of rational inattention to estimate individual differences in the capacity for cognitive control. *Proceedings of the Annual Meeting of the Cognitive Science Society, 44*.

Ham, H.*, Cedegao, Z.*, & Holliday, W. H. (2021). Does amy know ben knows you know your cards? a computational model of higher-order epistemic reasoning. *Proceedings of the Annual Meeting of the Cognitive Science Society, 43*.

Presentations

Automated Discovery of Cognitive Collaboration Dynamics.
06/08/2026 Poster given at Computational Cognitive Neuroscience Conference, New York, NY

Adaptivity of Collaborative Visual Working Memory.
02/08/2026 Talk given at Metacognitive Science Meeting, New York, NY

Rational Teachers Lie to Bounder Learners.
23/07/2026 Talk given at Annual Conference of the Cognitive Science Society, Rio de Janeiro, Brazil

Toward a Computational Cognitive Science of Collectives.
- 19/06/2026 Poster at Society for Philosophy and Psychology Annual Meeting, Johns Hopkins University
- 26/06/2026 Talk at Artificial Intelligence in Collectives Workshop, University of Tokyo, Japan

Collaborative encoding of visual working memory.
- 13/08/2025 Talk given at Computational Cognitive Neuroscience Conference, Amsterdam, Netherlands
- 01/08/2025 Poster presented at Annual Conference of the Cognitive Science Society, San Francisco, CA

A computational model of lexical false memory based on semantic distance from word embeddings.
28/07/2023 Talk given at Annual Conference of the Cognitive Science Society, Sydney, Australia

Effects of social context on reward-based learning.
- 01/10/2022 Poster presented at The Society for Neuroeconomics Conference, Arlington, VA
- 24/07/2026 Poster presented at Annual Conference of the Cognitive Science Society, Rio de Janeiro, Brazil

Leveraging psychometrics of rational inattention to estimate individual differences in the capacity for cognitive control.

- 19/05/2022 ◦ Talk given at Control Processes Conference, Online
- 10/06/2022 ◦ Poster presented at The Multi-disciplinary Conference on Reinforcement Learning and Decision Making, Brown University

Social Context Shapes Value Representation during Learning.

- 06/08/2021 ◦ Talk given at International Interdisciplinary Computational Cognitive Science Summer School, Online
- 23/09/2021 ◦ Talk given at Computational Cognition Workshop, Online

Does Amy Know Ben Knows You Know Your Cards? A Computational Model of Higher-Order Epistemic Reasoning.

29/07/2021 Poster presented at Annual Conference of the Cognitive Science Society, Online

Working Memory Shapes State-Space for Policy Formation in Reinforcement Learning.

30/07/2020 Poster presented at Annual Conference of the Cognitive Science Society, Online

Professional Services

Review

2026 **Computational Cognitive Neuroscience Conference**, *Reviewed 3 papers*.

2026 **Cognitive Science Society Annual Meeting**, *Reviewed 7 papers*.

2025 **Proceedings of the National Academy of Sciences**, *Reviewed 1 paper*.

2025 **Computational Cognitive Neuroscience Conference**, *Reviewed 3 papers*.

2025 **Cognitive Science Society Annual Meeting**, *Reviewed 4 papers*.

2024 **Cognitive Science Society Annual Meeting**, *Reviewed 1 paper*.

2023 **Cognitive Science Society Annual Meeting**, *Reviewed 1 paper*.

2022 **Cognitive Science Society Annual Meeting**, *Reviewed 2 papers*.

Volunteering

2024-2027 **PDP Meetings for Neural and Cognitive Computational Modeling**, *Co-organizer with Dr. Caroline Jahn*, https://pdp-princeton.github.io/PDP-schedule/.

2023 **Cognitive Science Society Annual Meeting**, *Onsite Session Assistant*.

2022 **Cognitive Science Society Annual Meeting**, *Online Technical Chair*.

Awards and Fundings

2025 **Rematch Summer Research Fellowship**, *Princeton University*.

2025 **Graduate School Teaching Award**, *Princeton University*.

2023-2028 **Centennial Fellowship in the Natural Sciences and Engineering**, *Princeton University*.

Teaching/Mentoring Experience

Teaching Assistantship

2027 Spring **fMRI Decoding: Reading Minds Using Brain Scans**, *Prof. Kenneth Norman*.

2025 Fall **Computational Models of Cognition**, *Profs. Thomas Griffiths & Brenden Lake*.

2024 Fall **The Cognitive Neuroscience of Selective Attention**, *Prof. Timothy Buschman*.

Students Mentorship

2026-Now **Anika Mehrotra**, *Undergraduate RA*, Princeton.

2026-Now **Eniola Owotomo**, *Undergraduate RA*, Princeton.

2025 **Yuanning (Helen) Hui**, *Rematch summer RA*, Princeton.

2024 - 2025 **Miles Sugarman**, *Psychology thesis student*, Princeton.

2023 **Ray Liu**, *Jenkins Lab Undergraduate*, UPenn.

2019-2020 **Kian Golestaneh, Megha Krishnan, Pearl Lee, Ailinh Nguyen, Robin Stewart, Jocelyn Xie**, *Undergraduate Lab for Cognitive Science and Psychology*, UC Berkeley.